Anqi Tan

Ph.D. Student in Bioengineering · University of Illinois Urbana-Champaign

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EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Present
Advisor: Prof. Brian T. Cunningham

B.S. in Mechanical Engineering, University of Illinois Urbana-Champaign Aug 2018 – Jun 2022

B.S. in Mechanical Engineering, Zhejiang University (ZJU-UIUC Institute) Aug 2018 – Jun 2022

RESEARCH INTERESTS

Biosensing Devices · Point-of-Care Diagnostics · Portable Optical Readers & Instrumentation · Hardware–Software Integration for Biomedical Measurement · Plasmon-Enhanced Lateral Flow Tests · Real-Time Hormone Monitoring

RESEARCH EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Built an $87 pocket-size NIR fluorescence reader for plasmonic-fluor-tagged lateral flow strips end to end: 750 nm excitation and custom optical filtering, Raspberry Pi imaging hardware, and an Android control app (Kotlin); reads a strip in 20 s and agrees with a lab ImageQuant scanner (R^2 = 0.99, 3.2% CV). First-author manuscript in preparation.
- Wrote the reader's analysis pipeline (automated test/control-line detection, background correction, 4PL calibration) that converts strip images into hormone concentrations.
- Leading development of the portable fluorescence reader for a $4.87M DoD (PRMRP) project on point-of-care phenotyping of sepsis and ARDS, in collaboration with Stanford, WashU, and UCSF.
- Leading the Live Hormone project: a competitive LFA platform for real-time testosterone, cortisol, and progesterone monitoring; the salivary cortisol assay reaches ≤0.12 ng/mL, with saliva collected from 153 adults (oral presentation, APHA 2026).
- Contributing to plasmon-enhanced LFAs for infectious disease detection and to reviews in Chemical Reviews and Lab on a Chip.

Research Assistant Intern, Institute of Brain Science, Zhejiang Qiangnao Technology Co., Ltd. (BrainCo), Hangzhou May 2022 – Aug 2022

- Supported R&D on two products, sleep aid via acoustic and electrical stimulation and fatigue monitoring with multimodal EEG: reviewed the literature and analyzed competing products, recruited and coordinated test volunteers, and collected experimental data.

Undergraduate Research Leader, ZJU-UIUC Institute, Zhejiang University 2019 – 2021

- Led a national-level research training project on a re-localizable RCM mechanism for minimally invasive surgery (first-author paper, IEEE ICARM 2021).
- Led a team implementing camera tracking for a robot-manipulated endoscope; contributed to a 3D-printed concrete dragon boat (Structures20/ACEM20).

Senior Design (ECE 445, Team #29): Mimicry Stage Lighting Control System Feb 2022 – May 2022

- Mechanical design lead: two-axis light manipulator (MATLAB forward/inverse kinematics; Euler–Lagrange and Newton–Euler torque analysis) and a 3D-printed, laser-cut structure designed for assembly and portability (9.3 kg, 4.5 h battery life).
- A handheld ESP32 + MPU9250 joystick drives the light in real time over WiFi; a music-analysis module generates LED color sequences.

RESEARCH FUNDING PARTICIPATION

DoD Peer Reviewed Medical Research Program, Award HT9425-25-1-0825 ($4.87M) 2025 – Present

- Point-of-care molecular phenotyping of sepsis and ARDS (PI: Pratik Sinha; with S. Singamaneni, C. Calfee, and B. T. Cunningham). Role: lead developer of the portable fluorescence reader.

PUBLICATIONS

1. **Anqi Tan**, Yuxiong Liu, Siun Lee, Han Keun Lee, Kai-Ning Wang, Tasfia Ashrafee, Ami Patel, Saurabh Umrao, Srikanth Singamaneni, Jenna L. Riis, Jacinda K. Dariotis, and Brian T. Cunningham. "LumaPF, a Lateral-Flow Unscanned Membrane Analyzer for Plasmonic Fluor: Real-Time Imaging of Competitive Assay Kinetics for Quantitative Salivary Cortisol." *Manuscript in preparation.*
2. **Anqi Tan**, M. Nguyen, T.-D. Doan, W.-C. Chen, T. Ayupova, H. T. N. Nguyen, S. Shepherd, and Brian T. Cunningham. "Quantification-Aware Active Learning for Expert-Efficient Particle Counting across Biosensing Imaging Platforms." *Manuscript in preparation.*
3. **Anqi Tan**, Seemesh Bhaskar, M. Nguyen, T.-D. Doan, Skye Shepherd, and Brian T. Cunningham. "Readout-Matched Plasmonic Cryosorets: Linker-Free Gold and Silver Nano-Assemblies as Lateral Flow Labels." *Manuscript in preparation.*
4. **Anqi Tan**, Kai-Ning Wang, Tasfia Ashrafee, Ami Patel, Yuxiong Liu, Saurabh Umrao, Srikanth Singamaneni, Jenna L. Riis, Jacinda K. Dariotis, and Brian T. Cunningham. "Portable Plasmonic Fluor-Enhanced Lateral Flow Assay for Quantitative Salivary Cortisol Measurement in Community Settings." *APHA 2026 Annual Meeting and Expo, San Antonio, TX. Accepted for oral presentation, Nov 3, 2026.*
5. Seemesh Bhaskar, Saurabh Umrao, Han Keun Lee, Joseph Tibbs, Amanda Bacon, Skye Shepherd, Takhmina Ayupova, Fatma Uysal Ciloglu, Leyang Liu, **Anqi Tan**, Wang-Chien Chen, My Thi Tra Nguyen, Maria Grace Scannell, Ugur Aygun, Ugur Parlatan, Catherine Zhang, Manish Kohli, Guy R. Adami, Wali Badar, Ron C. Gaba, Aaron Mansfield, Joel Schwartz, Xing Wang, Utkan Demirci, and Brian T. Cunningham. "Point of Care Molecular Cancer Diagnostics." *Lab on a Chip, 26(10), 2948–3001. 2026.*
6. Yuxiong Liu, Qisheng Jiang, **Anqi Tan**, Gayathri Kondepati, Prashant Gupta, Yixuan Wang, Yajur Chandar, Tracey H. Bach, Stacey L. House, Peter A. Rice, Bijal A. Parikh, Brian T. Cunningham, Sanjay Ram, Ige A. George, Srikanth Singamaneni, and Sumanth Gandra. "Ultrasensitive Point-of-Care Antigen Test for Detection of Neisseria gonorrhoeae Using Plasmon-Enhanced Lateral Flow Assays." *ACS Nano Medicine, 1(4), 1036–1047. 2026.*
7. Seemesh Bhaskar, Weijing Wang, Hankeun Lee, Leyang Liu, Saurabh Umrao, Weinan Liu, Amanda Bacon, Joseph Tibbs, Kodchakorn Khemtonglang, **Anqi Tan**, Takhmina Ayupova, Wang-Chien Chen, Xing Wang, and Brian T. Cunningham. "Photonic Crystal Grating Resonance and Interfaces for Health Diagnostic Technologies." *Chemical Reviews, 125(14), 6435–6540. 2025.*
8. Seemesh Bhaskar, Yanyu Xiong, Skye Shepherd, Joseph Tibbs, Amanda K. Bacon, Weinan Liu, Lucas D. Akin, Takhmina Ayupova, Weijing Wang, Hankeun Lee, Leyang Liu, **Anqi Tan**, Kodchakorn Khemtonglang, Xiaojing Wang, and Brian T. Cunningham. "Nano-Engineering at Functional Photonic Crystal Interfaces." *In Nano-Engineering at Functional Interfaces for Multi-Disciplinary Applications, Elsevier. 2025.*
9. Amanda Bacon, Weijing Wang, Hankeun Lee, Saurabh Umrao, Prima Dewi Sinawang, Demir Akin, Kodchakorn Khemtonglang, **Anqi Tan**, Sabina Hirshfield, Utkan Demirci, Xing Wang, and Brian T. Cunningham. "Review of HIV Self Testing Technologies and Promising Approaches for the Next Generation." *Biosensors, 13(2), 298. 2023.*
10. **Anqi Tan**, Shenwei Chen, Hanyu Gan, and Liangjing Yang. "Design Optimization of a Mechanically Constrained Re-Localizable Remote-Center-of-Motion." *2021 6th IEEE International Conference on Advanced Robotics and Mechatronics (ICARM). 2021.*
11. Y.Q. Feng, K.H. Zhang, **A.Q. Tan**, K. Ma, and Y. Xiao. "Utilization of 3D Printing for Concrete Dragon Boat Model." *The 2020 Structures Congress (Structures20), in the 2020 World Congress on Advances in Civil, Environmental, & Materials Research (ACEM20), Seoul, Korea, Aug 25–28, 2020.*

CONFERENCE PRESENTATIONS

1. Oral presentation (accepted): "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." APHA 2026 Annual Meeting and Expo, San Antonio, TX, Nov 3, 2026. (Nov 2026)
2. Poster: "A portable plasmonic fluor-enhanced lateral flow platform for quantitative salivary cortisol measurement." Lab-on-a-Chip and Microfluidics World Congress 2026 (SelectBIO), Laguna Hills, CA. (Oct 2026)
3. Poster: "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." Health Care Engineering Systems Symposium 2026 (Jump ARCHES), Peoria, IL. (Oct 2026)
4. Poster (co-author): "Assessing the acceptability and feasibility of real-time salivary hormone feedback and implications for building trust in biomedical research: A mixed-methods study." Health Care Engineering Systems Symposium 2026 (Jump ARCHES), Peoria, IL. (Oct 2026)
5. Poster: "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." 2026 Personalized Nutrition Innovation Day, UIUC. (Sep 2026)
6. Poster (co-author): "Assessing the acceptability and feasibility of real-time salivary hormone feedback and implications for building trust in biomedical research: A mixed-methods study." 2026 Personalized Nutrition Innovation Day, UIUC. (Sep 2026)
7. Poster: "Validating a lateral flow assay (LFA) scanning device for real-time salivary cortisol assaying." 2025 Personalized Nutrition Innovation Day, UIUC. (Oct 2025)

HONORS AND AWARDS

First Place ($3,000), Ashby Prize in Computational Science Hackathon, NCSA, UIUC	2025
Honorable Mention, Ashby Prize in Computational Science Hackathon, NCSA, UIUC	2024
First Prize, RoboMaster Mecha Master (College Level)	2019
Outstanding Performance and Innovation Award, 1st International Concrete Dragon Boat Competition (ICDBC), American Concrete Institute & Zhejiang University	2019
Third-Class Scholarship, Zhejiang University	2019
Public Service Pacesetter, Zhejiang University	2019
Model of Social Work, Zhejiang University (twice)	2019, 2020
Second Place, Social Research Competition, as team leader (three times)	2019 – 2020
Second Prize, Zhejiang Province Biology Olympiad	2017

TEACHING

Teaching Assistant, BIOE/ECE 415 Biomedical Instrumentation Lab, UIUC	Fall 2025
Teaching Assistant, ME 170 Computer-Aided Design, UIUC	Spring 2020, Spring 2021, Spring 2022

LEADERSHIP AND SERVICE

President, Residential College Student Committee, Zhejiang University, Haining Campus	2019 – 2022
Volunteer Teacher, Rural Education Program, Guizhou Province	Aug 2019

SKILLS

Biosensing & Wet Lab: Lateral Flow Assays, Plasmonic-Fluor Labels, ELISA, 3D Cell Culture, Fluorescence Staining

Mechanical Design & CAD: SolidWorks, Creo, Inventor, Fusion 360, AutoCAD, Blender

Fabrication & Simulation: 3D Printing, Melt Electrowriting, COMSOL, FEA

Programming & Hardware: Python, MATLAB, LabVIEW, Arduino, Android App Development (Kotlin)

Languages: English (Fluent), Chinese (Native)